

① (a) & (c) are LTI systems.

(b) This is not LTI, this system fails linearity:

$$x_1 \rightarrow \boxed{T} \rightarrow y_1 \quad x_2 \rightarrow \boxed{T} \rightarrow y_2$$

$$x_1 + x_2 \rightarrow \boxed{T} \rightarrow ??$$

$$\text{Output for input } x_1 + x_2 \text{ is } 1 + \int_{-\infty}^t x_1(\tau) d\tau + \int_{-\infty}^t x_2(\tau) d\tau$$

$$\text{But, } y_1 + y_2 = 2 + \int_{-\infty}^t x_1(\tau) d\tau + \int_{-\infty}^t x_2(\tau) d\tau$$

$$\neq 1 + \int_{-\infty}^t x_1(\tau) d\tau + \int_{-\infty}^t x_2(\tau) d\tau.$$

(d) This system fails time invariance, and hence is not LTI.

$$x \rightarrow \boxed{D} \xrightarrow{v} \boxed{T} \rightarrow w$$

$$x \rightarrow \boxed{T} \xrightarrow{z} \boxed{D} \rightarrow y$$

$$v(t) = x(t - t_0), \quad w(t) = t \cdot v(t) = t \cdot x(t - t_0)$$

$$z(t) = t \cdot x(t), \quad y(t) = z(t - t_0) = (t - t_0) \cdot x(t - t_0)$$

$$\therefore w \neq y.$$

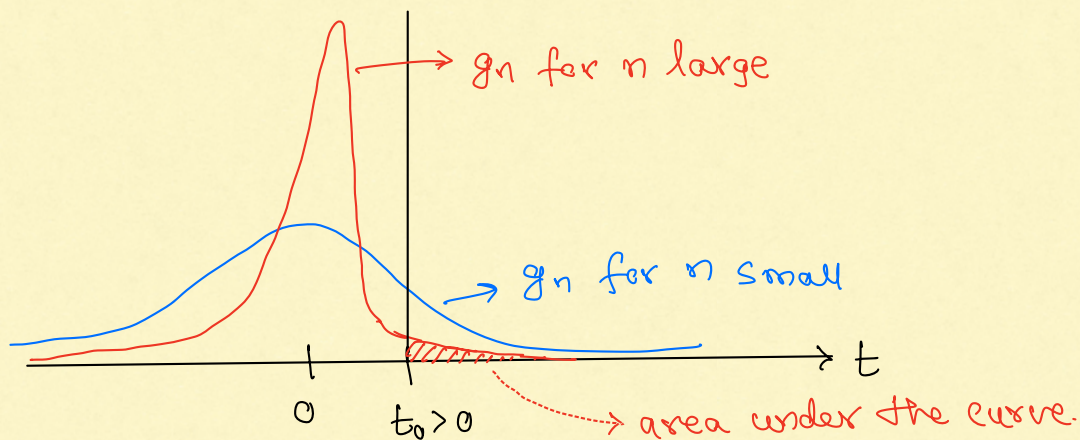
$\therefore$ The system is not time invariant.

②a

$$\int_{t_0}^{\infty} \delta(t) dt = \int_{-\infty}^{+\infty} \delta(t) u(t - t_0) dt$$

$$= \lim_{n \rightarrow \infty} \left[\int_{-\infty}^{+\infty} g_n(t) u(t - t_0) dt \right]$$

$$= \lim_{n \rightarrow \infty} \left[\int_{t_0}^{\infty} g_n(t) dt \right].$$



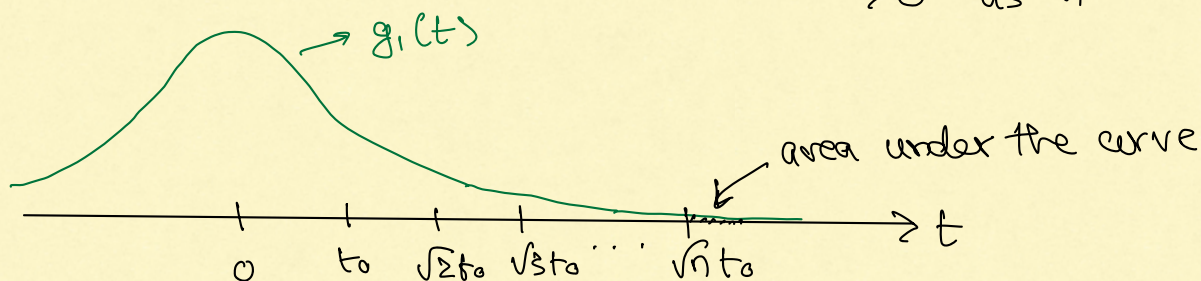
Consider $\int_{t_0}^{\infty} g_n(t) dt = \int_{t_0}^{\infty} \sqrt{\frac{n}{2\pi}} \cdot \exp\left(-\frac{nt^2}{2}\right) dt$

change of variables: $\tau = \sqrt{n} t$

$$\int_{t_0}^{\infty} g_n(t) dt = \int_{\sqrt{n} t_0}^{\infty} \sqrt{\frac{n}{2\pi}} \cdot \exp\left(-\frac{\tau^2}{2}\right) \frac{d\tau}{\sqrt{n}}$$

$$= \int_{\sqrt{n} t_0}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{\tau^2}{2}\right) d\tau = \int_{\sqrt{n} t_0}^{\infty} g_1(t) dt.$$

$\rightarrow 0$ as $n \rightarrow \infty$.



$$\therefore \lim_{n \rightarrow \infty} \left[\int_{t_0}^{\infty} g_n(t) dt \right] = 0 \text{ if } t_0 > 0.$$

Remark: To be precise, one can use the following known upper bound on the tail area of Gaussian curve.

$$\text{for } \gamma > 0: \quad 0 \leq \int_{\gamma}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt \leq \exp\left(-\frac{\gamma^2}{2}\right)$$

$$\therefore \int_{\gamma}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt \rightarrow 0 \quad \text{as } \gamma \rightarrow +\infty.$$

$$\therefore \int_{-\sqrt{n}t_0}^{\infty} g_1(t) dt \leq \exp\left(-\frac{nt_0^2}{2}\right)$$

Since $\exp\left(-\frac{nt_0^2}{2}\right) \rightarrow 0$ as $n \rightarrow \infty$, we arrive at the desired result.

2b

$$\int_{-\infty}^{t_0} \delta(t) dt = \lim_{n \rightarrow \infty} \int_{-\infty}^{t_0} g_n(t) dt$$

$$= \lim_{n \rightarrow \infty} \int_{-\infty}^{t_0} \frac{\sqrt{n}}{\sqrt{2\pi}} \exp\left(-\frac{nt^2}{2}\right) dt$$

$$= \lim_{n \rightarrow \infty} \int_{-\infty}^{\sqrt{n}t_0} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt, \quad \text{after change of variables: } \sqrt{n}t \rightarrow t.$$

(i) $t_0 < 0$:

By symmetry of the Gaussian curve

$$\int_{-\infty}^{\sqrt{n}t_0} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt = \int_{-\sqrt{n}t_0}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt \quad \text{where } -t_0 > 0.$$

$$\text{From question 2a: } \lim_{n \rightarrow \infty} \left[\int_{-\sqrt{n}t_0}^{+\infty} g_1(t) dt \right] = 0$$

(ii) $t_0 = 0$:

$$\int_{-\infty}^0 \delta(t) dt = \lim_{n \rightarrow \infty} \left[\int_{-\infty}^0 g_n(t) dt \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{2} \right] = \frac{1}{2}.$$

(iii) $t_0 > 0$:

$$\int_{-\infty}^{t_0} \delta(t) dt = \lim_{n \rightarrow \infty} \left[\int_{-\infty}^{t_0} g_n(t) dt \right]$$

$$= \lim_{n \rightarrow \infty} \left[1 - \int_{t_0}^{\infty} g_n(t) dt \right]$$

$$\left(\text{since } \int_{-\infty}^{t_0} g_n(t) dt + \int_{t_0}^{\infty} g_n(t) dt = \int_{-\infty}^{\infty} g_n(t) dt = 1 \right)$$

$$= 1 - \lim_{n \rightarrow \infty} \int_{t_0}^{\infty} g_n(t) dt$$

$$= 1 - 0 \quad (\text{from question 2a}).$$

$$= 1$$

2c

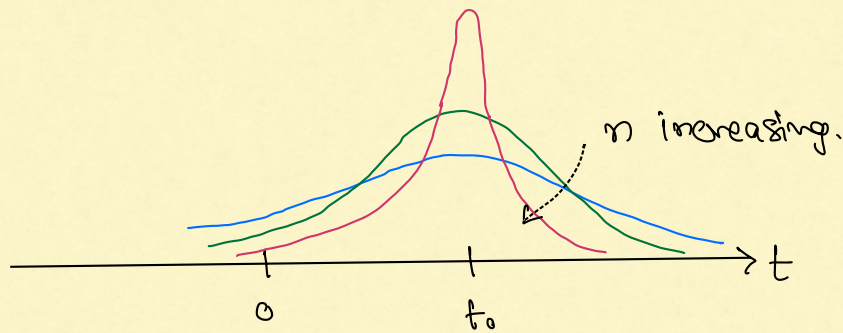
From Part 2b we knew:

$$\int_{-\infty}^t \delta(\tau) d\tau = \begin{cases} 1 & \text{if } t > 0 \\ 1/2 & \text{if } t = 0 \\ 0 & \text{if } t < 0 \end{cases} = u(t).$$

3a

$$\delta : g_1(t), g_2(t), \dots$$

$$\delta(t - t_0) : g_1(t - t_0), g_2(t - t_0), \dots$$



3b We are asked to use the sequence of functions.

$$\int_{-\infty}^{+\infty} \delta(t-t_0) \chi(t) dt = \lim_{n \rightarrow \infty} \left[\int_{-\infty}^{+\infty} g_n(t-t_0) \chi(t) dt \right]$$

$$= \lim_{n \rightarrow \infty} \left[\int_{-\infty}^{+\infty} g_n(t) \chi(t+t_0) dt \right]$$

(change of variables: $t \rightarrow t+t_0$)

$$= \int_{-\infty}^{+\infty} \delta(t) \chi(t+t_0) dt$$

= value of $\chi(t+t_0)$ at time $t=0$

$$= \chi(t_0).$$

3c
$$\int_{-\infty}^{+\infty} \delta(\tau-t_0) u(t-\tau) d\tau$$

$$= \int_{-\infty}^{+\infty} \delta(\tau) u(t-\tau-t_0) d\tau \quad (\text{using time-shifting formula})$$

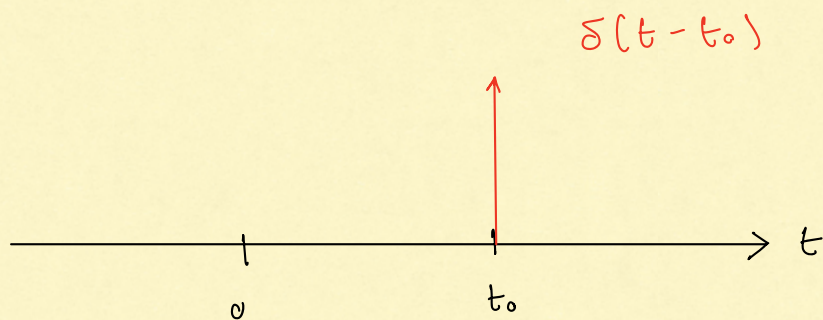
= value of $u(t-\tau-t_0)$ at time $\tau=0$

important result.

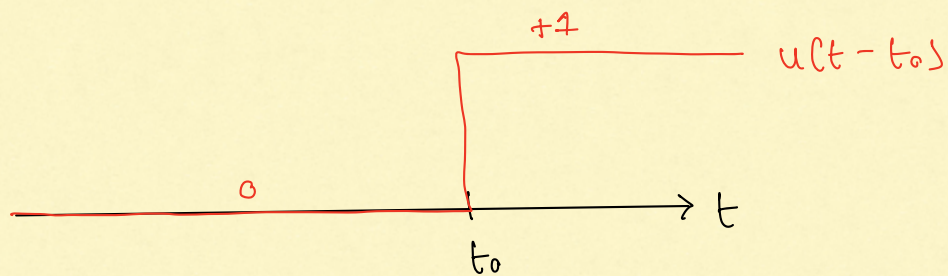
Memorize this.

$$\int_{-\infty}^t \delta(\tau-t_0) d\tau = u(t-t_0) = \begin{cases} 1 & \text{if } t > t_0 \\ 1/2 & \text{if } t = t_0 \\ 0 & \text{if } t < t_0 \end{cases}$$

$$\int_{-\infty}^t \delta(\tau - t_0) d\tau = u(t - t_0) = \begin{cases} 1 & \text{if } t > t_0 \\ 1/2 & \text{if } t = t_0 \\ 0 & \text{if } t < t_0 \end{cases}$$



$\Downarrow$ integrate



(3d)

We need to find t_0 such that

$$\langle \text{III}(t), x \rangle = \langle \text{III}(t-t_0), x \rangle \text{ for all } x.$$

$$\langle \text{III}(t), x \rangle = \int_{-\infty}^{+\infty} \text{III}(t) \cdot x(t) dt = \sum_{k=-\infty}^{+\infty} x(k) \quad \rightarrow (1)$$

$$\langle \text{III}(t-t_0), x \rangle = \int_{-\infty}^{+\infty} \text{III}(t-t_0) x(t) dt$$

$$= \int_{-\infty}^{+\infty} \text{III}(t) \cdot x(t+t_0) dt \quad (\text{from time-shift formula})$$

$$= \sum_{k=-\infty}^{+\infty} (\text{value of } x(t+t_0) \text{ at time } t=k)$$

$$= \sum_{k=-\infty}^{+\infty} x(k+t_0). \quad \rightarrow (2)$$

If t_0 is an integer then (1) and (2) are identical.

$$\sum_{k=-\infty}^{+\infty} x(k+t_0) = \sum_{m=-\infty}^{+\infty} x(m) \quad \text{where } m = k+t_0 \text{ is integer}$$

(4b)

$$\int_{-\infty}^{+\infty} (\delta(\tau+1) - \delta(\tau-1)) u(t-\tau) d\tau$$

$$= \int_{-\infty}^{+\infty} \delta(\tau+1) u(t-\tau) d\tau - \int_{-\infty}^{+\infty} \delta(\tau-1) u(t-\tau) d\tau$$

use time-shift formula:

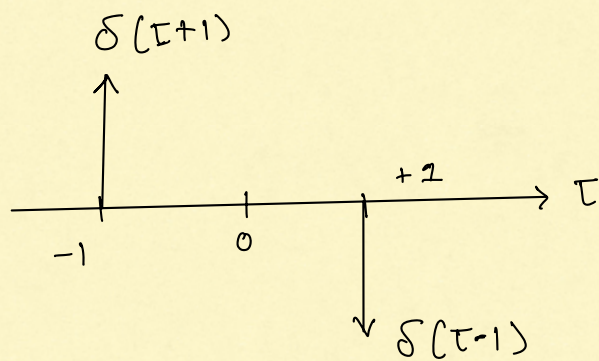
$$= \int_{-\infty}^{+\infty} \delta(\tau) u(t-\tau+1) d\tau - \int_{-\infty}^{+\infty} \delta(\tau) u(t-\tau-1) d\tau$$

$$= u(t+1) - u(t-1).$$

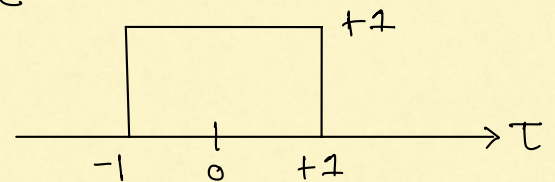
$$= \begin{cases} 0 & \text{if } t < -1 \\ 1 & \text{if } -1 < t < +1 \\ 0 & \text{if } t > 1. \end{cases}$$

Alternatively, we can use 3c : $\int_{-\infty}^t \delta(\tau - t_0) d\tau = u(t - t_0).$

$$\int_{-\infty}^t \delta(\tau + 1) d\tau - \int_{-\infty}^t \delta(\tau - 1) d\tau = u(t + 1) - u(t - 1).$$



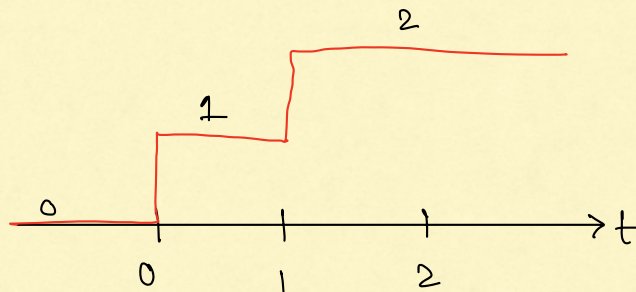
integrate
→



4c

Use the fact $\int_{-\infty}^t \delta(\tau - t_0) d\tau = u(t - t_0).$

We will obtain: $u(t) + u(t-1) + u(t-2)$



5e

$$\int_{-\infty}^{+\infty} \text{III}\left(\frac{t-1}{2}\right) x(t) dt$$

$$= \int_{-\infty}^{+\infty} \text{III}\left(\frac{t}{2}\right) x(t+1) dt$$

(time-shift formula:
 $t \rightarrow t+1$)

$$= 2 \int_{-\infty}^{+\infty} \text{III}(t) x(2t+1) dt$$

(time scaling formula
with $a = \frac{1}{2}$)

$$= 2 \cdot \sum_{k=-\infty}^{+\infty} x(2k+1)$$

(sum of $x(2t+1)$ at all
integer values of t)

5f

Yes, $\delta(-t) = \delta(t)$.

Prove this by showing

$$\langle \delta(-t), x \rangle = \langle \delta(t), x \rangle$$

for all x .

5g

$$\int_{-\infty}^{+\infty} \delta(2-t) x(t) dt$$

$$= \int_{-\infty}^{+\infty} \delta(2+t) x(-t) dt$$

(time reflection formula)
($t \rightarrow -t$)

$$= \int_{-\infty}^{+\infty} \delta(t) x(-(t-2)) dt$$

(time shift formula:
 $t \rightarrow t-2$)

$$= \text{value of } x(-(t-2)) \text{ at } t=0$$

$$= x(2).$$